



Investigation of office workers' sitting behaviors in an ergonomically adjusted workstation

Somayeh Tahernejad, Alireza Choobineh, Mohsen Razeghi, Mohammad Abdoli-Eramaki, Hossein Parsaei, Hadi Daneshmandi & Mozhgan Seif

To cite this article: Somayeh Tahernejad, Alireza Choobineh, Mohsen Razeghi, Mohammad Abdoli-Eramaki, Hossein Parsaei, Hadi Daneshmandi & Mozhgan Seif (2021): Investigation of office workers' sitting behaviors in an ergonomically adjusted workstation, International Journal of Occupational Safety and Ergonomics, DOI: [10.1080/10803548.2021.1990581](https://doi.org/10.1080/10803548.2021.1990581)

To link to this article: <https://doi.org/10.1080/10803548.2021.1990581>



Published online: 23 Nov 2021.



Submit your article to this journal [↗](#)



Article views: 337



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)



Investigation of office workers' sitting behaviors in an ergonomically adjusted workstation

Somayeh Tahernejad^a, Alireza Choobineh^{id a}, Mohsen Razeghi^{id c}, Mohammad Abdoli-Eramaki^d, Hossein Parsaei^{id e}, Hadi Daneshmandi^{id b} and Mozhgan Seif^{id f}

^aDepartment of Ergonomics, Shiraz University of Medical Sciences, Iran; ^bInstitute of Health, Shiraz University of Medical Sciences, Iran;

^cSchool of Rehabilitation Sciences, Shiraz University of Medical Sciences, Iran; ^dSchool of Occupational and Public Health, Ryerson University, Canada; ^eSchool of Medicine, Shiraz University of Medical Sciences, Iran; ^fSchool of Health, Shiraz University of Medical Sciences, Iran

ABSTRACT

Objectives. Common ergonomic office workstations are designed for a few optimum postures. Nonetheless, sitting is a dynamic activity and the ideal sitting posture is rarely maintained in practice. Therefore, the present study aimed to investigate the sitting behavior of office workers in an actual working environment using ergonomically adjusted workstations to examine whether they promote maintaining appropriate sitting postures. **Methods.** Sitting behaviors (frequency of postures and position changes in different body parts) were explored among 26 office workers during a 60-min sitting duration, using the posture recording and classification method developed by Graf et al. The rapid upper limb assessment (RULA) method was also used to assess postural load. Then, the results of the RULA method were compared with the results from investigating the sitting behavior of office workers. **Results.** Common ergonomic workstations were effective in eliminating some awkward postures. However, some important risk factors such as holding postures with an inappropriate lumbar spine curve (86% of the observations) and maintaining a posture for a long time (for 7–12 min) were observed in the participants' sitting behaviors, while they were neglected in the RULA method. **Conclusions.** The common ergonomic workstations could not guarantee the users' appropriate sitting behaviors.

KEYWORDS

office workers; sitting behavior; dynamic sitting; posture; prolonged sitting; rapid upper limb assessment

1. Introduction

Sitting behavior is defined as postures and positions of body parts held by an individual while sitting, and describes his/her postural range and movement frequencies [1]. Poor sitting behavior can cause musculoskeletal injuries [2]. Using ergonomic risk assessment tools, such as rapid upper limb assessment (RULA), and creating workstations in accordance with ergonomic guidelines and standards are common measures for office workers to reduce musculoskeletal disorders (MSDs) [3,4]. However, there are some ambiguities regarding the effectiveness of such guidelines, which have received less attention in the literature. As a widely accepted belief among ergonomists, the office workers who score high in conventional posture assessment methods (e.g., RULA) and have workstations that conform to common ergonomic standards are in acceptable working conditions, needing no corrective action. In the present semi-experimental quantitative field study, the validity of this belief is verified by investigating the sitting behaviors of office workers and comparing the results with the findings of the RULA method.

A review of the literature on ergonomics for office work and sitting behaviors provides the framework of the objectives in this research.

1.1. Ergonomics, musculoskeletal disorders and office work

MSDs resulting from cumulative trauma are amongst the most common occupational disorders all around the globe. In

addition to their considerable effects on health, MSDs impose major financial burdens on both staff and industries [5,6]. During the last decades, the number of office workers has increased in different countries. Studies have indicated that 20–60% of office workers suffer from MSDs [7].

Today, the ergonomic standards and principles for office work are based on the results of previous studies. Although numerous studies have been conducted on the ergonomic design of office workstations, many questions have remained unanswered [8]. As an instance, the majority of ergonomic investigations have been focused on anthropometric and biomechanical methods. In anthropometric techniques, body dimensions are measured in a standard sitting erect posture, while this posture is rarely adopted in practice. Therefore, such an ergonomic design might not be appropriate in an actual working environment. In biomechanical assessments, the spine posture, muscle contraction or pressure distribution is measured. However, sitting is a dynamic activity, so it is necessary to go beyond sitting biomechanics and take into consideration the nature of working behavior in sitting positions [8,9]. Up to now, a definition has not been given for an appropriate adolescent sitting posture [10,11]. Yet dynamic sitting behavior including frequent postural changes has been more popular compared to maintaining one 'ideal' posture [1,8,12]. In the workplace, however, chairs put individuals in one or two ideal postures, and ergonomic evaluations are done based on maintaining an ideal posture [13]. This can be attributed to several reasons, which have not received much attention in recent investigations.

To date, various ergonomic evaluations and interventions have been carried out to determine and reduce the incidence rate of MSDs among office workers [14]. However, there are ambiguities in their effectiveness because the ergonomic investigations performed in actual working environments normally use available observational tools, such as the posture classification techniques, for assessing the impacts of ergonomic interventions [1,15]. These techniques are not usually appropriate for the assessment of sitting behaviors because they have been developed for designing multipurpose systems and basically for evaluating heavy manual tasks [1]. The RULA method is normally used for the ergonomic evaluation of office work since it is the most common method for ergonomic evaluation of the tasks involving upper limbs [3,16]. Rapid entire body assessment (REBA) [17,18], rapid office strain assessment (ROSA) [4], posture checklists [19,20] and measurement of trunk and neck flexion and extension angles [21,22] have been used in some studies. Nevertheless, in these ergonomic evaluation methods, office tasks are assessed without taking into account the differences between office work and other jobs (specific risk factors of working in a sitting position).

It is believed that the correct sitting posture necessitates a straight back with knees and hips forming right angles [16]. Indeed, some studies have proposed other sitting postures, such as reclined and declined sitting postures, for office workers at computer workstations [16]. However, it should also be noted that a posture maintained for a prolonged period can have adverse effects on health [3,16,17,22]. The proper sitting behavior involves frequent changes in the positions of different body parts, which is known as 'dynamic sitting'. Therefore, the proper sitting behavior corresponds to the situations in which the office workers sit most of the time in optimal postures while considering the principle of dynamic sitting.

As mentioned, the results of common ergonomic assessments and using common chairs and ergonomic workstations in offices may not guarantee the proper sitting behavior of the workers.

1.2. Sitting behavior

Among the studies performed on sitting behavior with various objectives, a few studies have investigated the sitting behavior of office workers in actual working environments at ergonomically adjusted workstations.

To evaluate the sitting behavior, some studies have utilized observational methods and checklists categorizing sitting postures [13,23–26]. Numerous studies have also employed different technologies for the measurement of sitting behavior, including video analysis [27], accelerometers [28–30], motion analysis systems [12,31–33], power sensors [34,35], pressure distribution sensors [2,36,37] and Kinect software [38]. Most of these studies were carried out in laboratory environments. A few studies that have been carried out in actual working environments have not addressed a considerable number of important sitting postures. Among the observational methods used for the assessment of sitting behavior, the method proposed by Graf et al. seems to be more accurate in recording postures since it considers more aspects of sitting behavior [1]. In the present study, the sitting behavior of office workers was evaluated at computer workstations using the method developed by Graf et al. [13].

1.3. Study objectives

The objectives of this study are to determine the sitting behavior of office workers at ergonomically adjusted workstations, which use appropriate ergonomic features and adjustments. This study was conducted in an actual working environment with appropriate ergonomic workstations based on the ROSA method.

The study aims to answer the following questions:

- What are the common positions of different body parts at an ergonomically adjusted workstation during prolonged sitting?
- How often do the positions of different body parts of office workers change at ergonomically adjusted workstations during prolonged sitting?
- How do the results of the posture analysis using RULA compare with the posture classification method by Graf et al.?

Investigating the aforementioned issues is expected to provide a realistic insight into ergonomic evaluation and identification of the reasons for MSDs in office work.

2. Methods

2.1. Study setting and participants

This quantitative quasi-experimental field study aimed to examine the sitting behavior of office workers in the Municipal Organization of Shiraz, Iran. This cross-sectional study was conducted in April and May 2020. A total of 110 office workers were employed in this organization. All workstations were checked using the ROSA checklist [4]. The ROSA method was developed using the postures that were described in CSA Z412 guidelines on office ergonomics [4].

The inclusion criteria were a maximum ROSA score of two, a body mass index (BMI) between 18.5 and 24.9 [39], and having no apparent physical and mental problems (self-reported).

Among the eligible individuals, 26 (13 males and 13 females) were selected via simple random sampling. It should be noted that the workplaces of participants were not private and there were two to six people working in each room.

All subjects voluntarily participated in the study after receiving information about the study objectives. They also signed an informed consent form before the commencement of the study. Additionally, the study protocol was approved by the ethics committee of Shiraz University of Medical Sciences.

2.2. Data gathering tools

In this study, the required data were collected using the following tools/methods.

2.2.1. Demographic questionnaire

The questions in this survey covered demographic factors (including age, height, weight, marital status, education level and involvement in regular weekly sport/physical activities).

2.2.2. Posture classification system proposed by Graf et al.

In the present study, the sitting behavior of office workers was investigated at computer workstations using the classification system proposed by Graf et al. [13]. Figure 1 shows the posture classification proposed by Graf et al. This classification system particularly focuses on seated activities [13]. A matrix with 34

			1	2	3	4	5	6
	Spine	Trunk Position						
1								
2								
3								
4								
5								
6								

Figure 1. Posture classification system of Graf et al. [13]. Note: Side one of the sitting position classification matrix. The reverse side of the matrix is used when the shoulders are not parallel to the hip. Leg positions are numbered from 1 to 6 (columns).

positions was used for the shoulders in parallel with the hips and 34 positions for the twisted body. This system was used for the positions of the body and legs of the subjects with 1-min recording intervals for 1 or 2 h. To prevent complications due to the awareness of subjects about being observed, analysis was conducted only on the last 50 observations. First, the shoulder posture of the subjects (bending sideward or twisting relative to the hip) was recorded. Next, the lumbar spine posture of the subjects was determined in terms of exhibiting lordotic or kyphotic curves. The observers were asked to score a kyphotic posture for the lumbar spine noticeably flexed forward. Then, three distinct postures were considered, including forward inclination of the upper trunk relative to the hip, the upper trunk right above the hip and backward inclination of the upper trunk. The position of the legs was recorded as the final step [1].

2.2.3. Rapid upper limb assessment tool

In this study, the RULA method was employed to assess working postures. In this method, a score is assigned to each body part posture. The most neutral posture is assigned '1' and the most uncomfortable one is assigned '4'. The combination of the scores of the wrists, elbows and shoulders is denoted by 'A' and that of the legs, trunk and neck is denoted by 'B'. Muscle use in each task is assigned a score of '1', while the exerted force is assigned a score of '0' since these postures are static or very repetitive while requiring insignificant loading. These values are added to A and B to yield 'C' and 'D', respectively. The combination of C and D, also called the grand score, is in the range 1–7, and indicates the postural musculoskeletal loading. A low value of grand score (1 or 2) implies an acceptable working posture with action level 1. If the grand score is 3 or 4, a detailed examination is required and some changes are

probably necessary (action level 2). If the score is 5 or 6, prompt investigation and early modifications are needed (action level 3). Finally, if the grand score is 7, immediate investigation and modifications are necessary (action level 4) [3].

2.3. Procedure

2.3.1. Filming office workers at work

The office workers were filmed for 1 h while the participants were doing their routine office tasks. For each participant, one (two) camera(s) (one camera was used for filming the overall posture and another was needed for capturing the leg position) was (were) placed on a tripod(s). It is also worth mentioning that all videos were taken between 9:00 and 11:00. Additionally, the participants completed the demographic questionnaire.

2.3.2. Measuring sitting behavior

According to Graf et al. [13], a video frame was chosen for recording the posture with 1-min intervals (Figure 2). It should be noted that only the last 50 video frames were analyzed because the individuals' awareness of being filmed could affect their sitting behaviors during the first 10 min [13].

The postures were recorded based on the posture classification system proposed by Graf et al. [13]. The results were used to determine the frequency of various postures or positions of different body parts as well as the number of changes in positions of different parts during 50 min of sitting while doing office tasks (a change of position was recorded when the posture was altered from the previous recording).



Figure 2. Examples of images selected to record sitting postures of the participants.

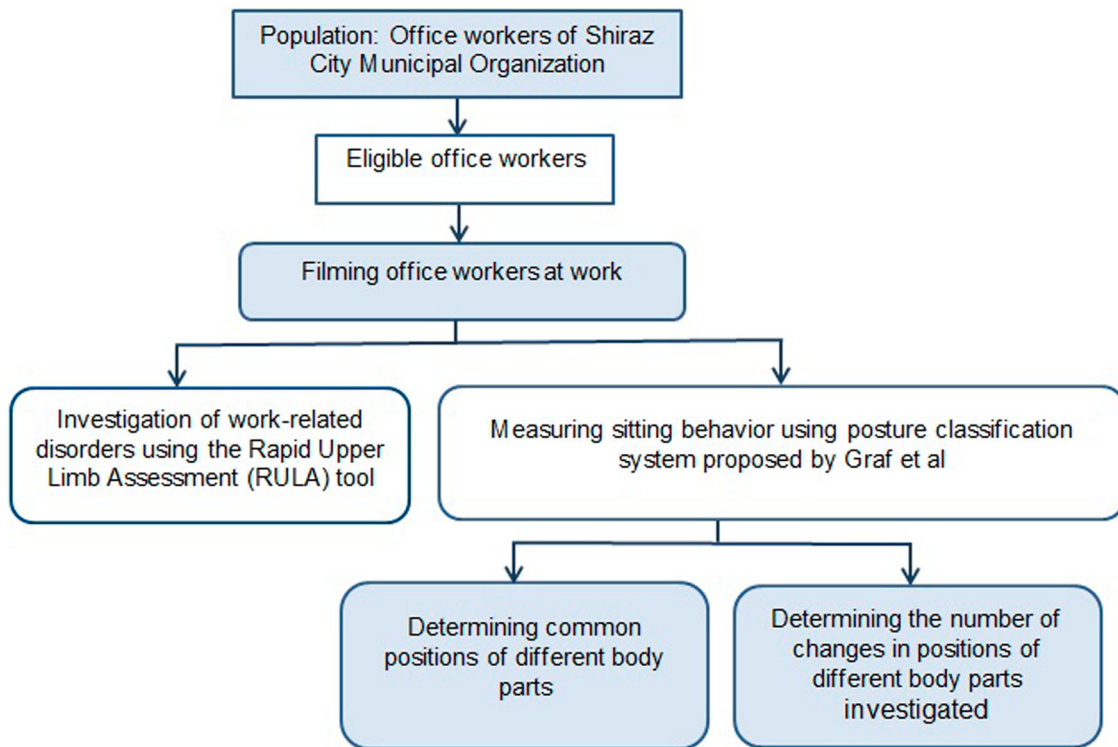


Figure 3. Flowchart of the study process.

2.3.3. Investigation of work-related disorders using the RULA method

The RULA method was used to assess the risk level of MSDs in the participants [3]. For determining the RULA score, video frames were sampled every 2 min resulting in 25 samples (starting at $t = 10$ and ending at $t = 60$). After calculating the RULA score for each video frame, the median RULA score was calculated for each participant to assess the MSD risk. To avoid inter-observer bias, the steps of the RULA method were performed by a trained (ergonomist) observer.

A flow diagram of the study process is shown in Figure 3.

2.4. Statistical analysis

The data were analyzed using Microsoft Excel version 12 and SPSS version 22.0. To answer the first question of the research, initially, the frequency percentage for each body part position in the study population was calculated and the results were plotted. Then, postures or positions that had a frequency greater than 20% were assumed to be the common ones. To answer the second question of the research, the mean and standard deviation of the variable 'number of changes' in postures or positions of each body part over 50 min of sitting were calculated for the subjects and the results were presented as a

table. The results of the RULA method, including the median, mean and standard deviation of the RULA scores and the frequency percentage of each action level, were calculated and presented as a table.

3. Results

3.1. Demographic characteristics

The participants' demographic characteristics are presented in Table 1. Accordingly, the age and the job experience mean $\pm$ standard deviation of the participants were 35.1 ± 3.6 and 8.3 ± 3.9 years, respectively.

3.2. Sitting behavior

The results indicated that the participants worked with a computer in 73% of the observations. The participants maintained a sitting posture in 98.23% of the observations, while they had left their workstations (standing or walking positions) in 1.77% of the observations.

Frequency percentages of different trunk positions, lumbar curve postures and leg positions are presented in Figures 4, 5 and 6, respectively. The analysis of these results can be

Table 1. Demographic characteristics of the studied office workers ($N = 26$).

Variable	Mean $\pm$ SD or n (%)
Quantitative	
Age (years)	35.1 $\pm$ 3.6
Job experience (years)	8.3 $\pm$ 3.9
Body mass index	23.9 $\pm$ 3.0
Qualitative variables	
Education level	
Bachelor	17 (65.3)
Master	9 (34.6)
Exercise	
Yes	10 (38.4)
No	16 (61.5)

summarized as follows: forward trunk position (50.1%), middle trunk position (33.8%), kyphotic lumbar curve (86.1%), neutral knee posture (right 54.2%) and knee flexion posture (left 53%) existed in more than 20% of the observations. Furthermore, these postures were considered as the common postures or positions of different body parts at an ergonomically adjusted workstation (answer to the first question of the research).

To answer the second question of the research, the number of changes in posture or positions of different body parts in the study population were recorded over 50 min of seated office work (Table 2). The results showed that the studied office workers rarely change their postures or positions of different body parts.

According to the results presented in Table 2, the position of the trunk relative to the hips changed 0.13 times per minute on average, i.e., one change every 7.7 min. In addition, the lumbar spine curve postures concerning kyphosis and lordosis changed 0.11 times per minute on average during 50 min of seated office work. The number of changes in feet positions was smaller compared to other body parts (Table 2).

3.3. The RULA results

The results of the RULA evaluation method demonstrated that 84.6 and 15.4% of the participants were in action levels 1 and 2, respectively. Accordingly, their postures while doing their tasks were acceptable or required further evaluations, which might necessitate modifications [3].

The mean grand score of RULA among the study population was 1.96. The mean scores of neck and trunk were 1.38 and 1.05, respectively, representing appropriate postures of these two regions. A score of '1' was assigned to all participants for the leg posture since in the RULA method a score of one is considered in the case that the feet are appropriately supported and the weight is equally balanced on both of them. The mean scores of lower arms and upper arms were 1.26 and 1.19, respectively.

4. Discussion

The present study aimed to provide detailed insight into the sitting behavior of office workers in an actual work setting using an ergonomically adjusted workstation.

4.1. Postures and positions of body parts

Maintaining improper postures for long periods is a characteristic of incorrect sitting behavior. In the present study, first, the frequency of postures or positions of different body parts at an ergonomically adjusted workstation was investigated and common positions were determined (answer to the first question of the research). In the following, the results of this part of the study are discussed.

4.1.1. Trunk positions

The results indicated that the trunks of participants were inclined forward relative to the hips in almost half of the observations, while in one-third of the observations, the trunks were above the hips (middle). Participants rarely used backrests. Despite the availability of ergonomic chairs, their tasks (working with a computer 73% of the time) prevented them from leaning backward. Trunks were in lateral bending or twisted postures in a few observations. This was proper sitting behavior, which could appear because of the appropriate dimensions of the workplace.

In the same vein, Grandjean and Hünting [25] reported that visual display terminal (VDT) operators preferred a trunk flexion posture, while only 10% maintained an upright trunk posture. However, Graf et al. [13] studied VDT operators and reported the middle trunk position most of the time and kyphotic postures in less than 10% of the observations. Cantoni et al. [24] also argued that call center operators were

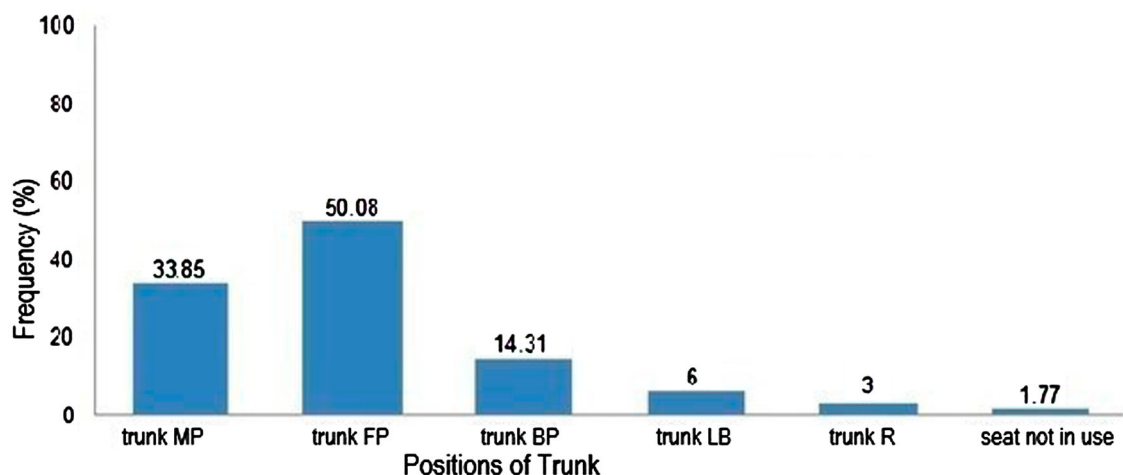


Figure 4. Frequency percentage of positions of the trunk. Note: trunk BP = backward trunk position; trunk FP = forward trunk position; trunk LB = trunk lateral bending; trunk MP = middle trunk position; trunk R = trunk rotation.

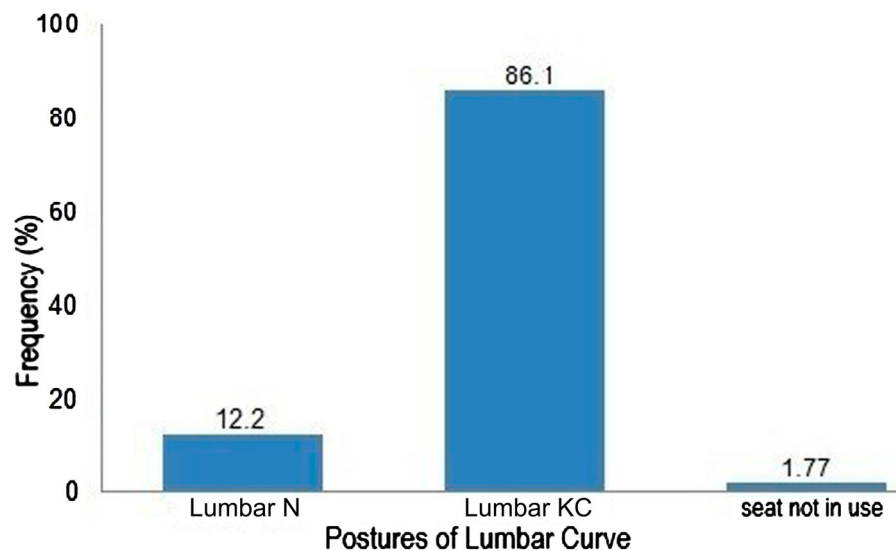


Figure 5. Frequency percentage of postures of the lumbar curve. Note: lumbar KC = kyphotic lumbar curve; lumbar NC = natural lumbar curve.

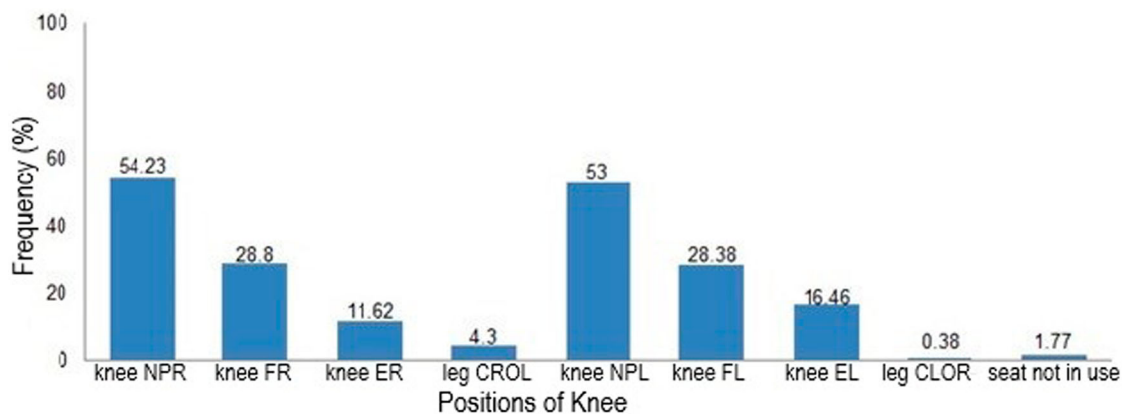


Figure 6. Frequency percentage of positions of the knee. Note: knee EL = left knee extension; knee ER = right knee extension; knee FL = left knee flexion; knee FR = right knee flexion; knee NPL = neutral left knee position; knee NPR = neutral right knee position; leg CROL = leg crossing right on left; leg CLOR = leg crossing left on right.

usually in the back-supported posture. These results contradict those of the present study; this can be attributed to the type of tasks and the number of participants. In the present study, the participants worked with computers in 73% of the observations and usually used the mouse. Dunk and Callaghan [31] also determined the highest and the lowest degrees of bending while typing and using the mouse, respectively. Furthermore, other studies have shown that operators bent their trunks forward for writing activities and general office tasks [13]. Similar results were also obtained by Barbieri et al. [29].

4.1.2. Postures of lumbar curve

The lumbar spine curve was in the kyphosis status in most observations. Most studies have reported a slumped posture as one of the most harmful sitting postures [40], which could increase the possibility of lower back pain (LBP) [27]. The findings of a study on call center operators indicated that the staff were usually in a moderately slumped posture [41]. In another study carried out by Mörl and Bradl [42], the lumbar spine was mostly flexed. These results are in agreement with those of the present study. The kyphotic lumbar curve posture is frequently

Table 2. Position change frequency for different body parts made during the sitting behavior measurement period (50 min).

Posture	N	Minimum–maximum number of changes	Mean number of changes	SD	Postural change frequency (change/min)
Trunk positions	26	3–14	6.77	2.74	0.13
Lumbar curve postures	26	2–12	5.62	2.75	0.11
Right leg positions	26	1–8	4.08	2.34	0.08
Left leg positions	26	0–8	3.85	2.33	0.07

Note: Trunk position change is a change of position recorded when the trunk posture (forward, middle or backward) was altered from the previous recording. Lumbar curve postural change is a change of posture recorded when the lumbar curve posture (natural lumbar curve or kyphotic lumbar curve) was altered from the previous recording. Leg position change is a change of position recorded when the leg posture (neutral knee, knee flexion or knee extension) was altered from the previous recording.

observed among office workers, and many studies and standards have encouraged maintaining the natural lumbar spine curve while sitting [8]. However, even in assessment methods based on ergonomic observations such as postural loading on the upper body assessment (LUBA) and RULA, which have the sitting posture included in their posture classification, the middle trunk position with kyphotic lumbar curve has not been mentioned. Moreover, no posture with incorrect lumbar spine curve has been considered in the posture classification of these methods [3,43].

4.1.3. Leg positions

The leg position in a sitting posture affects the knee angle and hamstring length, and consequently the pelvic position [12]. In the present study, the knees were in the neutral position (90°) in almost half of the observations; this position is appropriate for knees while sitting [8]. Interestingly, the participants often used the chair wheels as a footrest in order to put their knees in a flexed position. Leg crossing is also one of the components of sitting behavior. Graf et al. [13] reported leg crossing among the operators of VDT (30%). Intermittent loading and unloading of body weight on soft tissues and muscles and balance of lower extremities during prolonged sitting have been mentioned as the reasons for leg crossing [1]. However, leg crossing was observed only in a few cases in the present study.

4.2. Frequency of change in position

Another important feature of correct sitting behavior is related to changing the posture of different body parts while sitting. Therefore, in the second question of the research, the number of changes in positions of different body parts was investigated. The results showed that while performing office tasks, people rarely change their posture or positions of different body parts.

A posture maintained for a prolonged period can harm health [1,8]. Previous studies have reported insufficient nutrition of the intervertebral discs, increased height of the spine, decreased range of lumbar motion and increased lumbar spine stiffness as side effects of prolonged sitting [8]. Low and relatively static muscle activity during seated work can lead to muscle discomfort and pain [44].

In the current study, the highest postural changes were related to trunk postures, which were changed 6.77 times on average during 50 min of working in the sitting position. In a previous study on call center operators, the mean number of postural changes per working hour was nearly 12 among healthy people and 10 among those with different levels of LBP [37]. These values were higher in the present study, which seemed normal because postural changes were assessed separately in different body parts. Grandjean and Hünting [25] also reported that VDT operators moved rarely. However, Graf et al. [13] performed a study among public office workers and reported frequent changes in their body position.

In this study, the number of changes in leg position was smaller compared to other parts of the body. Prolonged sitting can lead to swelling of the lower extremities, while activation of the vein pump system by moving the legs can reduce foot and leg edema [45].

Based on ergonomic principles, individuals are recommended to maintain a single posture for at most 20% of their maximum holding time (MHT) [46]. Nonetheless, sufficient data are not available for estimating the MHT in different

positions of working in a sitting position. Grandjean quantifies static loading in three categories relating to the forces required. If a large force is exerted, the static muscle actions should be less than 10 s; for a moderate force it is less than 1 min and for a low force it is less than 4 min [3]. However, considering the frequency of postural changes in different body parts in the present study, the MHT of each posture was far beyond the ergonomic recommendations. Dynamic sitting behavior could provide beneficial biological and physiological effects because postural variations could reduce spinal loads [12,47]. It is also worth noting that ergonomic workstations that provide more comfortable positions may cause individuals to hold a certain posture for a longer time, causing pain and damage by another mechanism (buffer effect) [48]. Moreover, the results of this study revealed considerable differences among different body parts with respect to the frequency of postural changes, which was consistent with the results of other studies conducted on this issue [12].

4.3. Prolonged sitting

In the present study, the participants left their chairs only in 1.77% of the observations, indicating that they encountered the risk factors of prolonged sitting. Considering the static features of sitting, fatigue in cartilage and bone tissues, muscles, ligaments and the related tissues as well as insufficient restoration can result in health problems in the long run [48–52]. In a study by Mork and Westgaard [41], the participants maintained standing or walking postures in 5% of their working time, which was in agreement with the findings of the present study.

4.4. Comparison of the sitting behavior investigation findings to the RULA results

Monitoring the sitting behavior of participants revealed important risk factors, such as excessive slumped posture and low frequency of postural changes for all body parts. However, the RULA method revealed acceptable results (answer to the third question of the research). The posture assessment results from the RULA method are in agreement with the results obtained by Kar and Hedge, in which the RULA score of the participants who performed a computer typing task for 1 h at an ergonomically adjusted workstation was 1.683 (action level 1) [53].

The mean scores of the neck and trunk were 1.38 and 1.05, respectively, representing appropriate postures of these two regions. Although the trunk and neck were flexed or bent forward in a notable number of observations, their flexion angles were not largely based on the criteria of RULA and not considered harmful. In addition, the spinal curve is not evaluated by the RULA technique, while the studies comparing sitting and standing positions or different sitting postures have shown that the lumbar spine posture affects and is affected by many biomechanical and physiological variables [54–56]. Therefore, assessment of the lumbar spine curve while sitting is of great importance [32].

Another important point is that maintaining a static posture for more than 1 min adds one point to the score in the RULA method. Considering the results presented in Table 2, common postures were maintained for nearly 8 min for the trunk and nearly 12 min for the feet, but their harmfulness scores were equal to that for holding the posture for 1 min. Furthermore, a

large body of evidence has demonstrated that the sitting duration was associated with the onset of LBP [6,37]. The findings of the present study also indicated that prolonged sitting was one of the features of sitting behavior, but the total duration of sitting has not been addressed in the RULA method [57,58].

Finally, office work is quite different from other non-sitting occupations regarding the trunk, neck and leg postures (curves and angles), muscle activity level and static activity [25,40,59,60], but these differences have not been taken into account in common evaluation methods like RULA. Several risk factors of office work that are frequently detected in the sitting behavior of office workers have been neglected in RULA. Thus, the results of these evaluations might lead to the misconception that adjustment of workstations in accordance with ergonomic standards would prevent MSDs.

4.5. Limitations

Although the method presented by Graf et al. [13] is more accurate in recording postures compared to other techniques, precise postures of all body parts, including neck posture and location of the buttocks on the seat, cannot be recorded. Therefore, in this study, these postures were not recorded. However, it is clear that the postures of these parts may have different effects on the biomechanics of the body.

5. Conclusion

The results of this study indicated that forward trunk position, middle trunk position, kyphotic lumbar curve, neutral knee posture and knee flexion posture are the common postures or positions of different body parts that were maintained by the office workers at ergonomically adjusted workstations. In this study, the subjects rarely changed their posture or positions of different body parts at the workstations. As the results of this study suggest, providing an ergonomically adjusted workstation can reduce the risk factors of MSDs to some extent in the study population. However, still there exist some other important risk factors, such as prolonged slumped posture and violating the principle of dynamic sitting.

Nevertheless, the evaluation results from the RULA method were acceptable for the study population. Therefore, it is suggested that in future studies, besides using ergonomic evaluation methods for investigating the causes of inappropriate sitting behavior of office workers, these methods have also to be considered for spine curve and principle of dynamic sitting.

Acknowledgements

This article was extracted from a thesis written by Mrs Somaye Tahernejad, PhD student of ergonomics, and financially supported by Shiraz University of Medical Sciences.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was financially supported by Shiraz University of Medical Sciences [grant No. 98-01-04-20367].

ORCID

Alireza Choobineh  <http://orcid.org/0000-0002-0866-0404>

Mohsen Razeghi  <http://orcid.org/0000-0003-2179-851X>

Hossein Parsaei  <http://orcid.org/0000-0003-1052-8434>

Hadi Daneshmandi  <http://orcid.org/0000-0003-3505-9590>

Mozhgan Seif  <http://orcid.org/0000-0003-2301-5603>

References

- [1] Grieco A, Molteni G, De Vito G. Sitting behavior. In: Nico DD, Haslegrave DC, Chaffin DD, editors. Working postures and movements: tools for evaluation and engineering. Boca Raton (FL): CRC Press; 2004. p. 144–155.
- [2] Roossien C, Stegenga J, Hodselmans A, et al. Can a smart chair improve the sitting behavior of office workers? *Appl Ergon*. 2017;65: 355–361. doi:10.1016/j.apergo.2017.07.012
- [3] McAtamney L, Corlett EN. RULA: a survey method for the investigation of work-related upper limb disorders. *Appl Ergon*. 1993;24(2):91–99. doi:10.1016/0003-6870(93)90080-5
- [4] Sonne M, Villalta DL, Andrews DM. Development and evaluation of an office ergonomic risk checklist: ROSA – rapid office strain assessment. *Appl Ergon*. 2012;43(1):98–108. doi:10.1016/j.apergo.2011.03.008
- [5] Motabar H, Nimbarde AD, Raub E. Strength, endurance and fatigue response of rotator cuff muscles during isometric exertions. *Int J Ind Ergon*. 2019;71:128–135. doi:10.1016/j.ergon.2019.03.001
- [6] Besharati A, Daneshmandi H, Zareh K, et al. Work-related musculoskeletal problems and associated factors among office workers. *Int J Occup Saf Ergon*. 2020;26(3):632–638. doi:10.1080/10803548.2018.1501238
- [7] Hoe VC, Urquhart DM, Kelsall HL, et al. Ergonomic interventions for preventing work-related musculoskeletal disorders of the upper limb and neck among office workers. *Cochrane Database Syst Rev*. 2018;10(10):CD008570. doi: 10.1002/14651858.CD008570.pub3
- [8] Woo E, White P, Lai C. Ergonomics standards and guidelines for computer workstation design and the impact on users' health – a review. *Ergonomics*. 2016;59(3):464–475. doi:10.1080/00140139.2015.1076528
- [9] Springer T. The future of ergonomic office seating. New York: Knoll Workplace research, Knoll; 2010.
- [10] Claus AP, Hides JA, Moseley GL, et al. Is 'ideal' sitting posture real?: measurement of spinal curves in four sitting postures. *Man Ther*. 2009;14(4):404–408. doi:10.1016/j.math.2008.06.001
- [11] Robertson M, Amick IIIB, DeRango K, et al. The effects of an office ergonomics training and chair intervention on worker knowledge, behavior and musculoskeletal risk. *Appl. Ergon*. 2009;40(1):124–135. doi:10.1016/j.apergo.2007.12.009
- [12] Niekerk S-M, Louw QA, Grimmer-Sommers K. Frequency of postural changes during sitting whilst using a desktop computer – exploring an analytical methodology. *Ergonomics*. 2014;57(4):545–554. doi:10.1080/00140139.2014.884247
- [13] Graf M, Guggenbühl U, Krueger H. An assessment of seated activity and postures at five workplaces. *Int. J. Ind. Ergon*. 1995;15(2):81–90. doi:10.1016/0169-8141(94)00027-Z
- [14] Bazazan A, Dianat I, Feizollahi N, et al. Effect of a posture correction-based intervention on musculoskeletal symptoms and fatigue among control room operators. *Appl. Ergon*. 2019;76:12–19. doi:10.1016/j.apergo.2018.11.008
- [15] Lowe BD, Dempsey PG, Jones EM. Ergonomics assessment methods used by ergonomics professionals. *Appl Ergon*. 2019;81:102882. doi:10.1016/j.apergo.2019.102882
- [16] Swinton PA, Cooper K, Hancock E. Workplace interventions to improve sitting posture: a systematic review. *Prev Med*. 2017;101: 204–212. doi:10.1016/j.ypmed.2017.06.023
- [17] Al Madani D, Dababneh A. Rapid entire body assessment: a literature review. *Am J Eng Appl Sci*. 2016;9(1):107–118. doi:10.3844/ajeassp.2016.107.118
- [18] Hignett S, McAtamney L. Rapid entire body assessment (REBA). *Appl Ergon*. 2000;31(2):201–205. doi:10.1016/S0003-6870(99)00039-3
- [19] Culig KM, Dickinson AM, Lindstrom-Hazel D, et al. Combining workstation design and performance management to increase ergonomically correct computer typing postures. *J Organiz Behav Manage*. 2008;28(3):146–175. doi:10.1080/01608060802251064
- [20] Mahmud N, Kenny DT, Md Zein R, et al. The effects of office ergonomic training on musculoskeletal complaints, sickness absence, and psychological well-being: a cluster randomized control trial. *Asia Pacific J Public Health*. 2015;27(2): NP1652-68. doi:10.1177/1010539511419199

- [21] Meinert M, König M, Jaschinski W. Web-based office ergonomics intervention on work-related complaints: a field study. *Ergonomics*. 2013;56(11):1658–1668. doi:[10.1080/00140139.2013.835872](https://doi.org/10.1080/00140139.2013.835872)
- [22] Konarska M, Wolska A, Widerszal-Bazyl M, et al. The effect of an ergonomic intervention on musculoskeletal, psychosocial and visual strain of VDT data entry work: the Polish part of the international study. *Int J Occup Saf Ergon*. 2005;11(1):65–76. doi:[10.1080/10803548.2005.11076631](https://doi.org/10.1080/10803548.2005.11076631)
- [23] Branton P, Grayson G. An evaluation of train seats by observation of sitting behaviour. *Ergonomics*. 1967;10(1):35–51. doi:[10.1080/00140136708930838](https://doi.org/10.1080/00140136708930838)
- [24] Cantoni S, Colombini D, Occhipinti E, et al. Posture analysis and evaluation at the old and new work place of a telephone company. In: Grandjean E, editor. *Ergonomics and health in modern offices*. London: Taylor & Francis; 1984. p. 456–464.
- [25] Grandjean E, Hünting W. Ergonomics of posture – review of various problems of standing and sitting posture. *Appl Ergon*. 1977;8(3):135–140. doi:[10.1016/0003-6870\(77\)90002-3](https://doi.org/10.1016/0003-6870(77)90002-3)
- [26] Kamp I, Kilincsoy Ü, Vink P. Chosen postures during specific sitting activities. *Ergonomics*. 2011;54(11):1029–1042. doi:[10.1080/00140139.2011.618230](https://doi.org/10.1080/00140139.2011.618230)
- [27] Womersley L, May S. Sitting posture of subjects with postural backache. *J Manipulative Physiol Ther*. 2006;29(3):213–218.
- [28] Ryan CG, Dall PM, Granat MH, et al. Sitting patterns at work: objective measurement of adherence to current recommendations. *Ergonomics*. 2011;54(6):531–538. doi:[10.1080/00140139.2011.570458](https://doi.org/10.1080/00140139.2011.570458)
- [29] Barbieri DF, Srinivasan D, Mathiassen SE, et al. Variation in upper extremity, neck and trunk postures when performing computer work at a sit–stand station. *Appl Ergon*. 2019;75:120–128. doi:[10.1016/j.apergo.2018.09.012](https://doi.org/10.1016/j.apergo.2018.09.012)
- [30] Jia B, Nussbaum MA. Influences of continuous sitting and psychosocial stress on low back kinematics, kinetics, discomfort, and localized muscle fatigue during unsupported sitting activities. *Ergonomics*. 2018;61(12):1671–1684. doi:[10.1080/00140139.2018.1497815](https://doi.org/10.1080/00140139.2018.1497815)
- [31] Dunk NM, Callaghan JP. Gender-based differences in postural responses to seated exposures. *Clin Biomech (Bristol, Avon)*. 2005;20(10):1101–1110. doi:[10.1016/j.clinbiomech.2005.07.004](https://doi.org/10.1016/j.clinbiomech.2005.07.004)
- [32] Claus AP, Hides JA, Moseley GL, et al. Thoracic and lumbar posture behaviour in sitting tasks and standing: progressing the biomechanics from observations to measurements. *Appl Ergon*. 2016;53:161–168. doi:[10.1016/j.apergo.2015.09.006](https://doi.org/10.1016/j.apergo.2015.09.006)
- [33] Vergara M, Page A. Relationship between comfort and back posture and mobility in sitting-posture. *Appl Ergon*. 2002;33(1):1–8. doi:[10.1016/S0003-6870\(01\)00056-4](https://doi.org/10.1016/S0003-6870(01)00056-4)
- [34] Yamada M, Kamiya K, Kudo M, et al. Soft authentication and behavior analysis using a chair with sensors attached: hippint authentication. *Pattern Anal Appl*. 2009;12(3):251–260.
- [35] Zemp R, Tanadini M, Plüss S, et al. Application of machine learning approaches for classifying sitting posture based on force and acceleration sensors. *Biomed Res Int*. 2016;2016. doi:[10.1155/2016/5978489](https://doi.org/10.1155/2016/5978489)
- [36] Zemp R, Fliesser M, Wippert P-M, et al. Occupational sitting behaviour and its relationship with back pain – a pilot study. *Appl Ergon*. 2016;56:84–91. doi:[10.1016/j.apergo.2016.03.007](https://doi.org/10.1016/j.apergo.2016.03.007)
- [37] Bontrup C, Taylor WR, Fliesser M, et al. Low back pain and its relationship with sitting behaviour among sedentary office workers. *Appl Ergon*. 2019;81:102894. doi:[10.1016/j.apergo.2019.102894](https://doi.org/10.1016/j.apergo.2019.102894)
- [38] Sun X, Lan X, Zhou D, et al. Research on office chair based on modern office posture. In: Kurosu M, editor. *Proceedings of the 20th International Conference on Human–Computer Interaction*; 2018 Jul 15–20; Las Vegas (NV): Springer; 2018. p. 195–205.
- [39] Lee R, Nieman D. *Nutritional assessment*. New York (NY): McGraw-Hill; 2003.
- [40] Wong AY, Chan TP, Chau AW, et al. Do different sitting postures affect spinal biomechanics of asymptomatic individuals? *Gait Posture*. 2019;67:230–235. doi:[10.1016/j.gaitpost.2018.10.028](https://doi.org/10.1016/j.gaitpost.2018.10.028)
- [41] Mork PJ, Westgaard RH. Back posture and low back muscle activity in female computer workers: a field study. *Clin Biomech (Bristol, Avon)*. 2009;24(2):169–175. doi:[10.1016/j.clinbiomech.2008.11.001](https://doi.org/10.1016/j.clinbiomech.2008.11.001)
- [42] Mörl F, Bradl I. Lumbar posture and muscular activity while sitting during office work. *J Electromyogr Kinesiol*. 2013;23(2):362–368. doi:[10.1016/j.jelekin.2012.10.002](https://doi.org/10.1016/j.jelekin.2012.10.002)
- [43] Kee D, Karwowski W. LUBA: an assessment technique for postural loading on the upper body based on joint motion discomfort and maximum holding time. *Appl Ergon*. 2001;32(4):357–366. doi:[10.1016/S0003-6870\(01\)00006-0](https://doi.org/10.1016/S0003-6870(01)00006-0)
- [44] Van Dieën JH, Westebring-van der Putten EP, Kingma I, et al. Low-level activity of the trunk extensor muscles causes electromyographic manifestations of fatigue in absence of decreased oxygenation. *J Electromyogr Kinesiol*. 2009;19(3):398–406. doi:[10.1016/j.jelekin.2007.11.010](https://doi.org/10.1016/j.jelekin.2007.11.010)
- [45] Stranden E. Dynamic leg volume changes when sitting in a locked and free floating tilt office chair. *Ergonomics*. 2000;43(3):421–433. doi:[10.1080/001401300184503](https://doi.org/10.1080/001401300184503)
- [46] Miedema MC, Douwes M, Dul J. Recommended maximum holding times for prevention of discomfort of static standing postures. *Int J Ind Ergon*. 1997;19(1):9–18. doi:[10.1016/0169-8141\(95\)00037-2](https://doi.org/10.1016/0169-8141(95)00037-2)
- [47] Davis KG, Kotowski SE. Postural variability: an effective way to reduce musculoskeletal discomfort in office work. *Hum Factors*. 2014;56(7):1249–1261. doi:[10.1177/0018720814528003](https://doi.org/10.1177/0018720814528003)
- [48] Bendix T, Bridger B. Seating concepts. In: Nico DD, Haslegrave DC, Chaffin DD, editors. *Working postures and movements: tools for evaluation and engineering*. Boca Raton (FL): CRC Press; 2004. p. 156–167.
- [49] Thorp AA, Healy GN, Winkler E, et al. Prolonged sedentary time and physical activity in workplace and non-work contexts: a cross-sectional study of office, customer service and call centre employees. *Int J Behav Nutr Phys Act*. 2012;9(1):128. doi:[10.1186/1479-5868-9-128](https://doi.org/10.1186/1479-5868-9-128)
- [50] Hamberg-van Reenen HH, Van Der Beek AJ, Blatter BM, et al. Does musculoskeletal discomfort at work predict future musculoskeletal pain? *Ergonomics*. 2008;51(5):637–648. doi:[10.1080/00140130701743433](https://doi.org/10.1080/00140130701743433)
- [51] Kastelic K, Kozinc Ž, Šarabon N. Sitting and low back disorders: an overview of the most commonly suggested harmful mechanisms. *Coll Antropol*. 2018;42(1):73–79.
- [52] Daneshmandi H, Choobineh A, Ghaem H, et al. Proper sit–stand work schedule to reduce the negative outcomes of sedentary behavior: a randomized clinical trial. *Int J Occup Saf Ergon*. 2021 Apr 23: 1–17. doi:[10.1080/10803548.2019.1679972](https://doi.org/10.1080/10803548.2019.1679972)
- [53] Kar G, Hedge A. Effect of workstation configuration on musculoskeletal discomfort, productivity, postural risks, and perceived fatigue in a sit–stand–walk intervention for computer-based work. *Applied Ergonomics*. 2021 Jan 1;90:103211. doi:[10.1016/j.apergo.2020.103211](https://doi.org/10.1016/j.apergo.2020.103211)
- [54] Hedman TP, Fernie GR. Mechanical response of the lumbar spine to seated postural loads. *Spine*. 1997;22(7):734–743. doi:[10.1097/00007632-199704010-00004](https://doi.org/10.1097/00007632-199704010-00004)
- [55] Claus AP, Hides JA, Moseley GL, et al. Different ways to balance the spine: subtle changes in sagittal spinal curves affect regional muscle activity. *Spine*. 2009;34(6):E208–E214. doi:[10.1097/BR5.0b013e3181908ead](https://doi.org/10.1097/BR5.0b013e3181908ead)
- [56] Lee L-J, Chang AT, Coppieters MW, et al. Changes in sitting posture induce multiplanar changes in chest wall shape and motion with breathing. *Respir Physiol Neurobiol*. 2010;170(3):236–245.
- [57] Chen S-M, Liu M-F, Cook J, et al. Sedentary lifestyle as a risk factor for low back pain: a systematic review. *Int Arch Occup Environ Health*. 2009;82(7):797–806. doi:[10.1007/s00420-009-0410-0](https://doi.org/10.1007/s00420-009-0410-0)
- [58] Bakker EW, Verhagen AP, van Trijffel E, et al. Spinal mechanical load as a risk factor for low back pain: a systematic review of prospective cohort studies. *Spine*. 2009;34(8):E281–E293. doi:[10.1097/BR5.0b013e318195b257](https://doi.org/10.1097/BR5.0b013e318195b257)
- [59] Harrison DD, Harrison SO, Croft AC, et al. Sitting biomechanics part I: review of the literature. *JMPT*. 1999;22(9):594–609.
- [60] Bauer C, Rast F, Böck C, et al. Determination of a sagittal plane axis of rotation for a dynamic office chair. *Appl Ergon*. 2018;72:107–112. doi:[10.1016/j.apergo.2018.05.008](https://doi.org/10.1016/j.apergo.2018.05.008)